

2 *Parakaryon myojinensis* is a single-celled organism. The species was discovered in 2012.

P. myojinensis is unusual because it has features of prokaryotes and eukaryotes. It also has features that are **not** present in prokaryotes or in eukaryotes.

Figure 2.1 shows a diagram of *P. myojinensis*.

The endosymbionts present in the cell of *P. myojinensis* are organisms that can be present inside some prokaryotic cells and some eukaryotic cells.

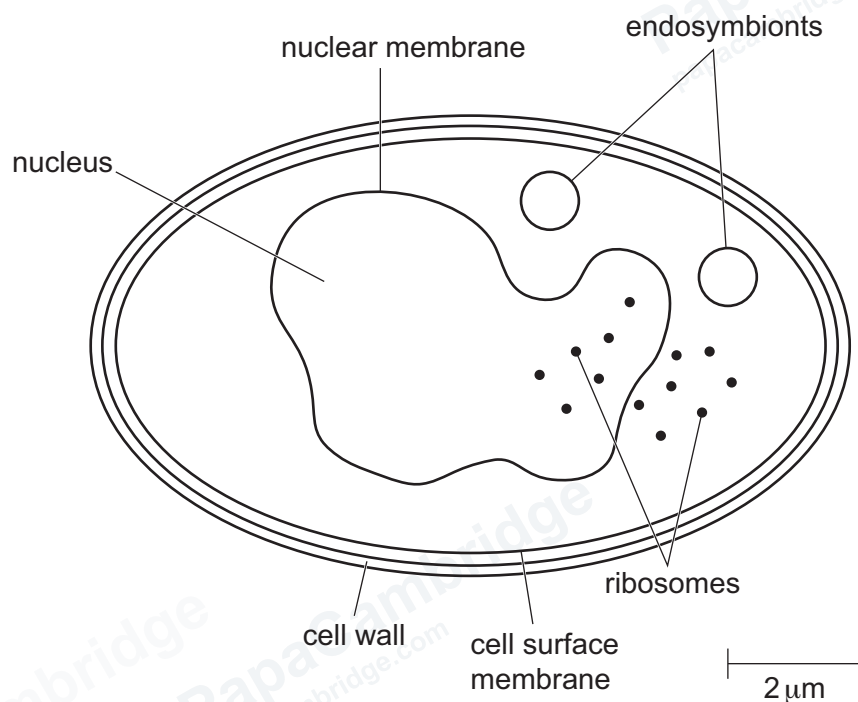


Figure 2.1

- (a) (i) State **one** difference, **other than** the presence of a nucleus and endosymbionts, between the *P. myojinensis* cell shown in Figure 2.1 and a typical prokaryotic cell.

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..... [1]

- (ii) State the differences between the *P. myojinensis* cell in Figure 2.1 and a typical eukaryotic cell.

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..... [2]



- (iii) Chemical analysis of the cell wall of *P. myojinensis* could provide information about how this organism compares with prokaryotes and with eukaryotes.

Suggest how the results of this chemical analysis may show how this organism compares with prokaryotes and with eukaryotes.

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..... [2]

- (b) Outline how the actual length of a cell of *P. myojinensis* can be determined using a light microscope, a stage micrometer and an eyepiece graticule.

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[Total: 7]